Spectrometer Experiment Report

Zheng Zhihao

Measurement of the Spectrometer's Field of View

Experiment Data

	Vernier1	Vernier2
Left	10.5	190.5
Right	8.48	188.48
$\Delta\varphi$	2.02	2.02
Average$\Delta\varphi$		2.02

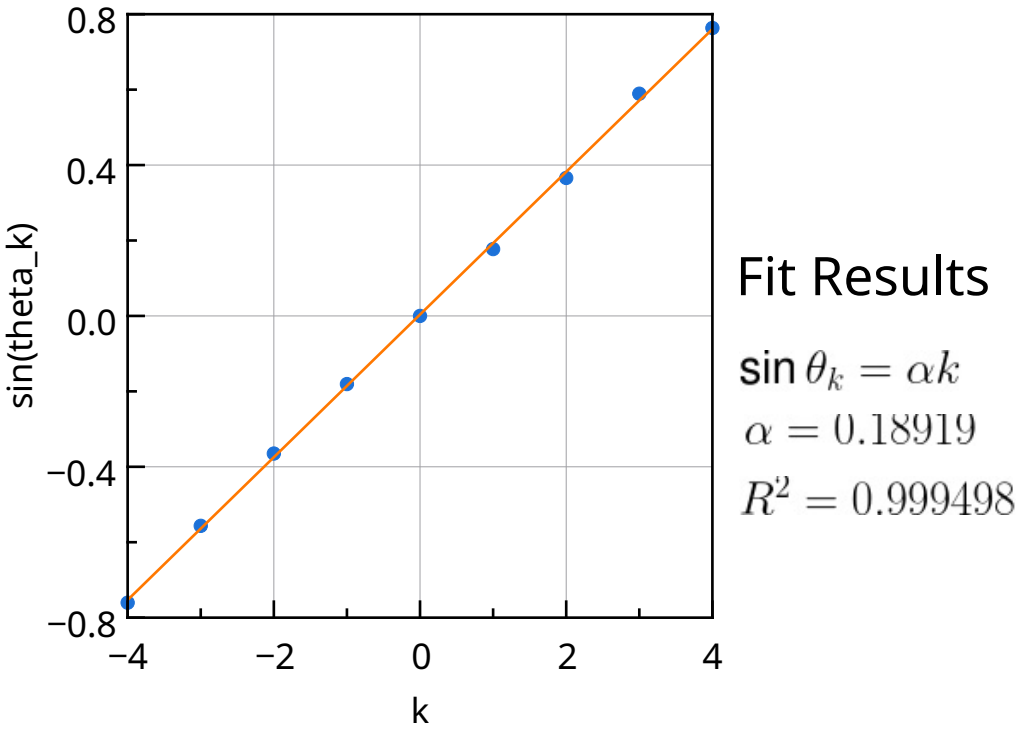
The average angular difference is $\overline{\Delta\varphi} = -2.02°$.

Measurement of the Sodium light Wavelength Using a Diffraction Grating

Experiment Data

k	Vernier1	Vernier2	θ_k		$sin\theta_k$	
0	210.27	30.27	Vernier1	Vernier2		
1	200.05	20.07	10.22	10.2	10.21	0.18
2	188.82	8.8	21.45	21.47	21.46	0.37
3	174.17	-5.83	36.1	36.1	36.1	0.59
4	160.67	-19.67	49.6	49.93	49.77	0.76
-1	220.68	40.67	10.42	10.4	10.41	-0.18
-2	231.67	51.67	21.4	21.4	21.4	-0.36
-3	244.08	64.08	33.82	33.82	33.82	-0.56
-4	259.75	79.77	49.48	49.5	49.49	-0.76

Linear Fit of $\sin\theta_k$ versus Diffraction Order k



The linear fit of $\sin\theta_k$ versus k yields $\alpha = 0.18919$ and $R^2 = 0.999498//$

$$\lambda = \alpha d = \frac{0.18919}{300} \times 10^6 = 630.63\,\text{nm}$$

Errors Analysis

The relative error is approximately 7.01%.Although the linear fitting yields a high coefficient of determination ($R^2 = 0.999498$), systematic errors may still exist. Possible sources of error include inaccurate angular readings, imperfect alignment of the spectrometer, and uncertainty in the grating constant.The experimental accuracy could be improved by carefully adjusting the optical system, measuring diffraction angles on both sides, and repeating the measurements.

Review Question

Ensure that the spectrometer is stable and properly maintained
Avoid disterbing the optical component after adjusting
Read both vernier scales to reduce eccentricity errors